

Exercise

`../data/gold_hist.txt` contains historical gold prices from 1972 to 2015

- Read the data as dataframe (price, year)

```
In [58]: # Your code here
# !cat ../data/gold_hist.txt
# import pandas as pd
# pd.read_csv('../data/gold_hist.txt')

f=open('../data/gold_hist.txt')
content=f.read()
f.close()
lines=content.split('\n')
years=[int(y) for y in lines[:2]]
prices=[float(l.strip().replace('$','').replace(',','')) for l in lines[1:222]]
```

```
In [59]: df=pd.DataFrame({'year':years, 'price':prices})
df.head()
```

```
Out[59]:
```

	year	price
0	2015	1060.00
1	1994	383.25
2	1973	106.48
3	2014	1199.25
4	1993	391.75

- Plot the data with time on the horizontal axis and gold price on the vertical axis (provide title, and labels for axes)

```
In [60]: # plt.scatter(df.year, df.price);
```

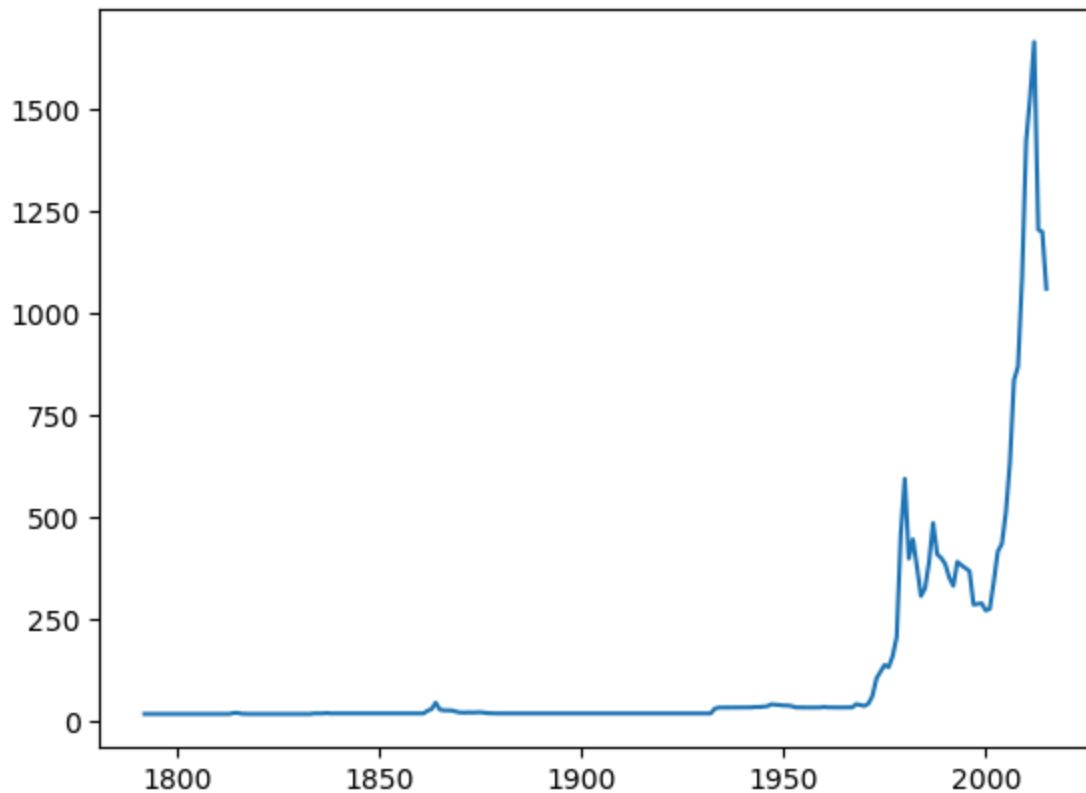
```
In [61]: df=df.sort_values(by='year')
```

```
In [62]: df.info()

<class 'pandas.core.frame.DataFrame'>
Index: 222 entries, 193 to 0
Data columns (total 2 columns):
#   Column  Non-Null Count  Dtype  
---  -
0   year    222 non-null     int64  
1   price   222 non-null     float64
dtypes: float64(1), int64(1)
memory usage: 5.2 KB
```

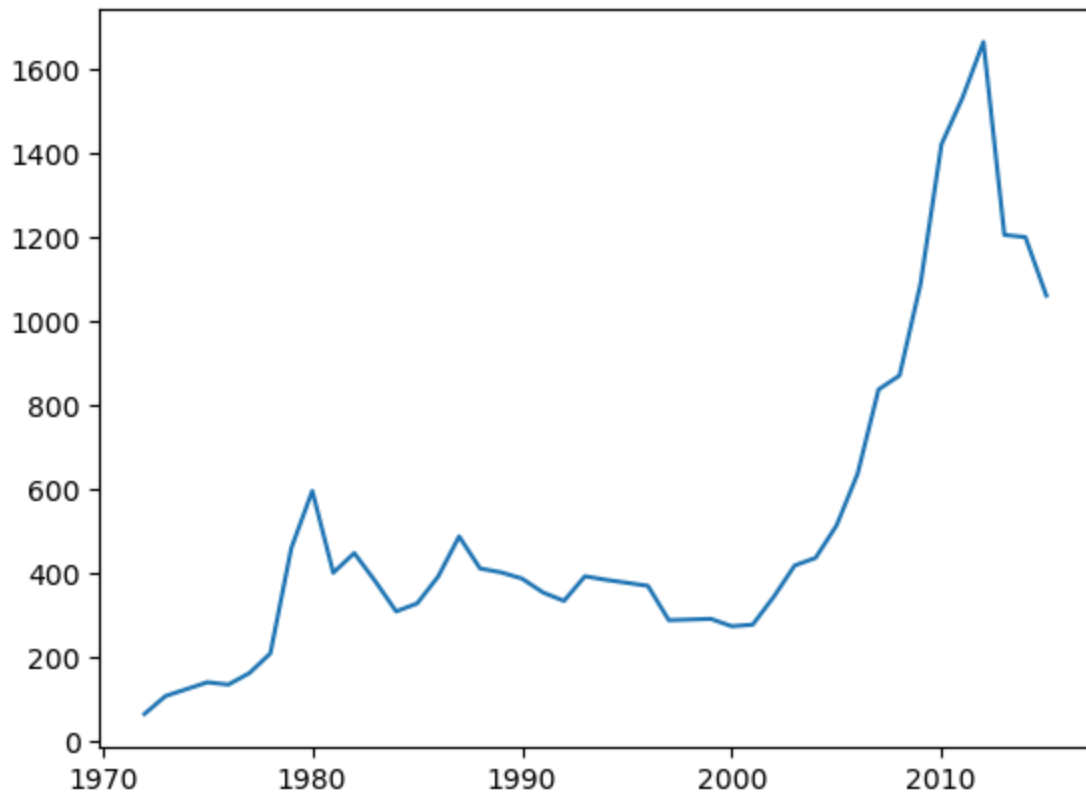
```
In [63]: # df.price= df.price.asfloat()
```

```
In [64]: import matplotlib.pyplot as plt  
plt.plot(df.year, df.price);
```



- Plot data again from 1972 and after
- Label the point that represents the peak price

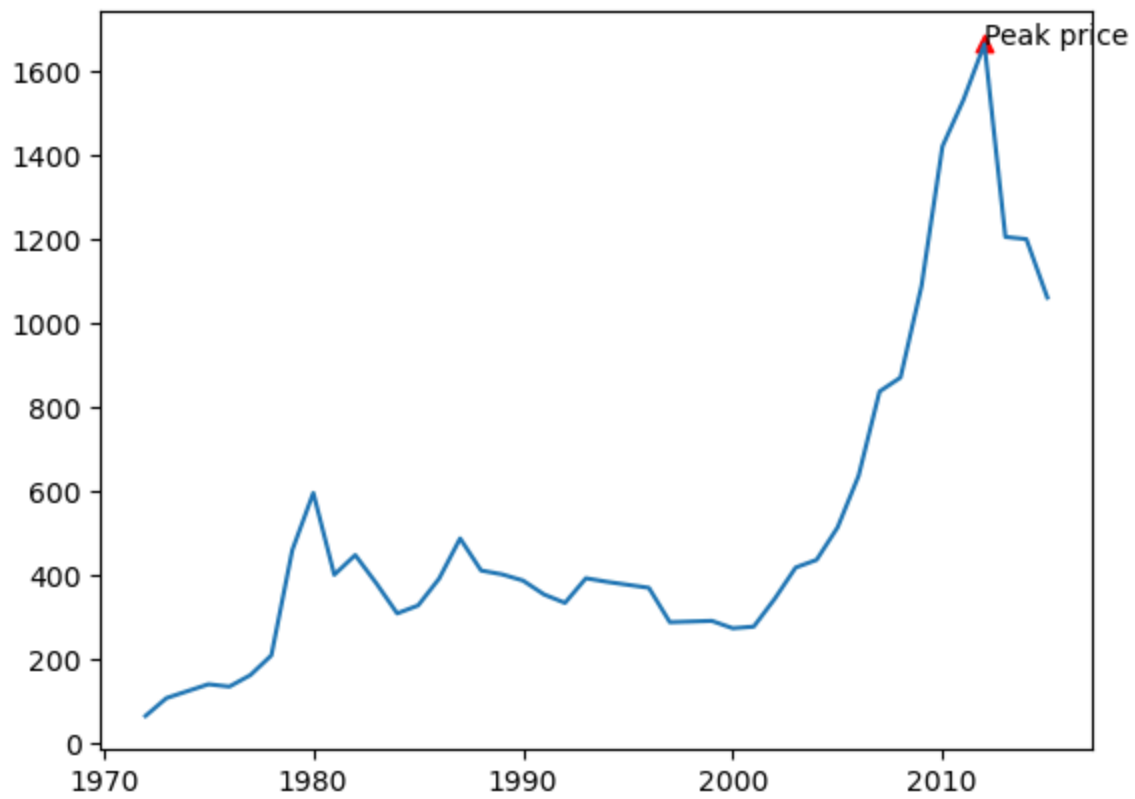
```
In [70]: # Your code here  
df=df.loc[df.year>=1972,:]  
# df.plot(x='year',y='price');  
plt.plot(df.year, df.price);
```



```
In [75]: peak_price=df.price.max()
peak_year= df.loc[df.price==df.price.max(),'year'].values[0]
```

```
In [80]: plt.plot(df.year, df.price)
plt.scatter(peak_year,peak_price, marker='^', color='red')
plt.text(peak_year,peak_price,'Peak price')
```

```
Out[80]: Text(2012, 1664.0, 'Peak price')
```



In []: